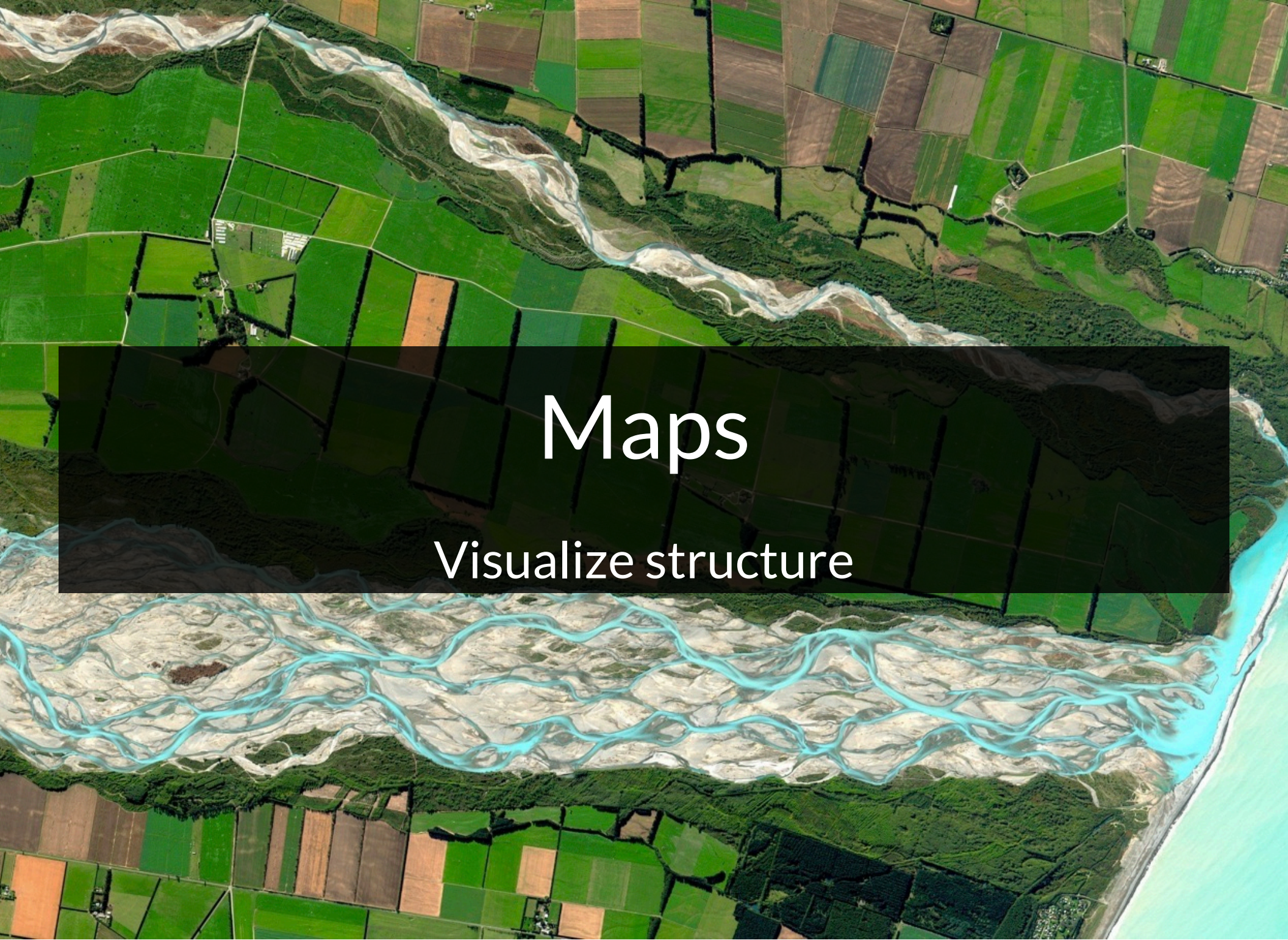
Chatbots as an Interface for Spatial Data and Analysis

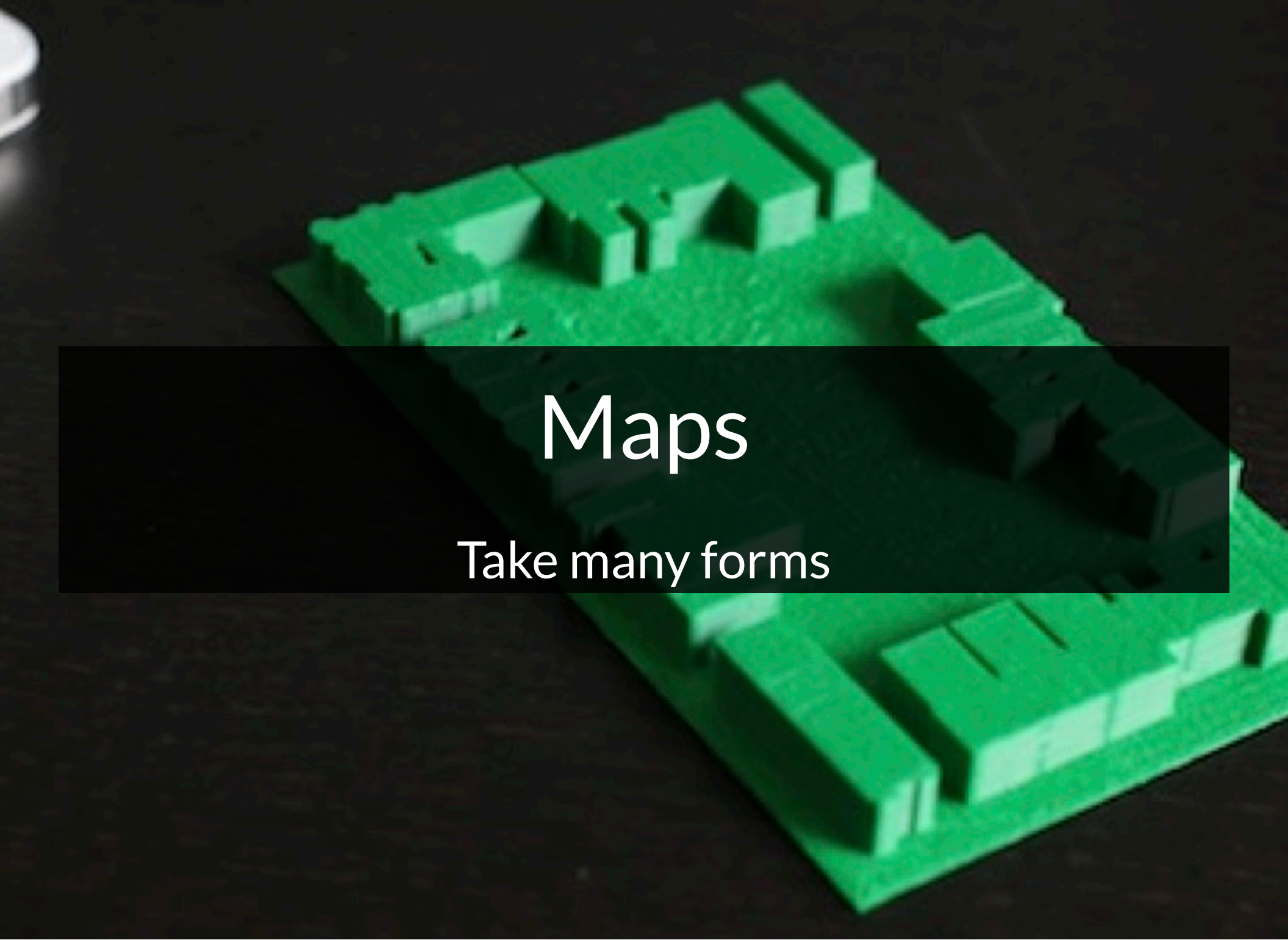
Andrew Turner / @ajturner

Director, Esri R&D DC



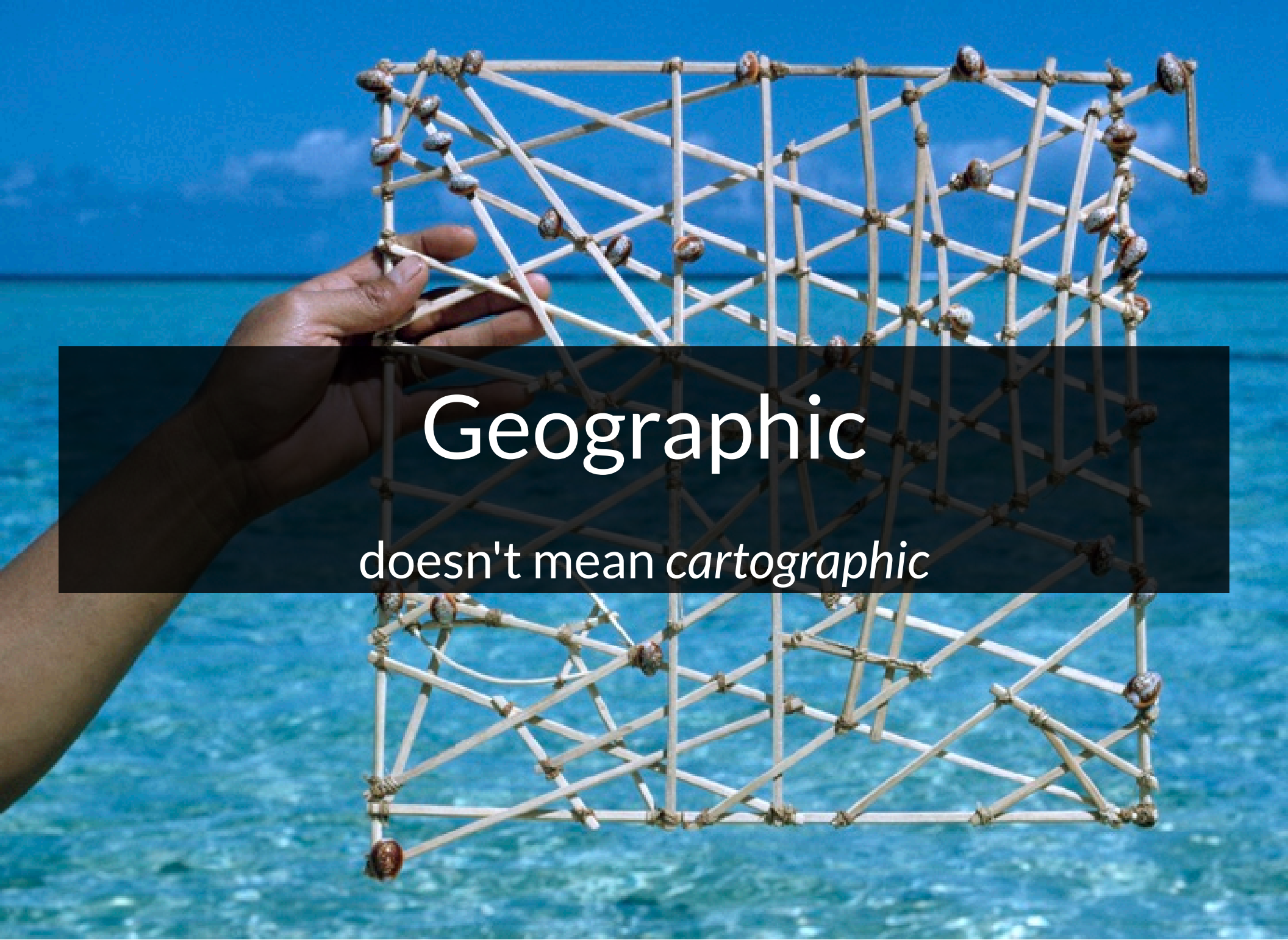
Maps

Visualize structure



Maps

Take many forms


Geographic
doesn't mean *cartographic*



Geographic Systems

Manage public, physical infrastructure


Infrastructure
Supports Community, and People
ent Laforet


People
have many perspectives



Data-Driven Citizenship


Public Digital Infrastructure



Accessible

Meet people where they are...

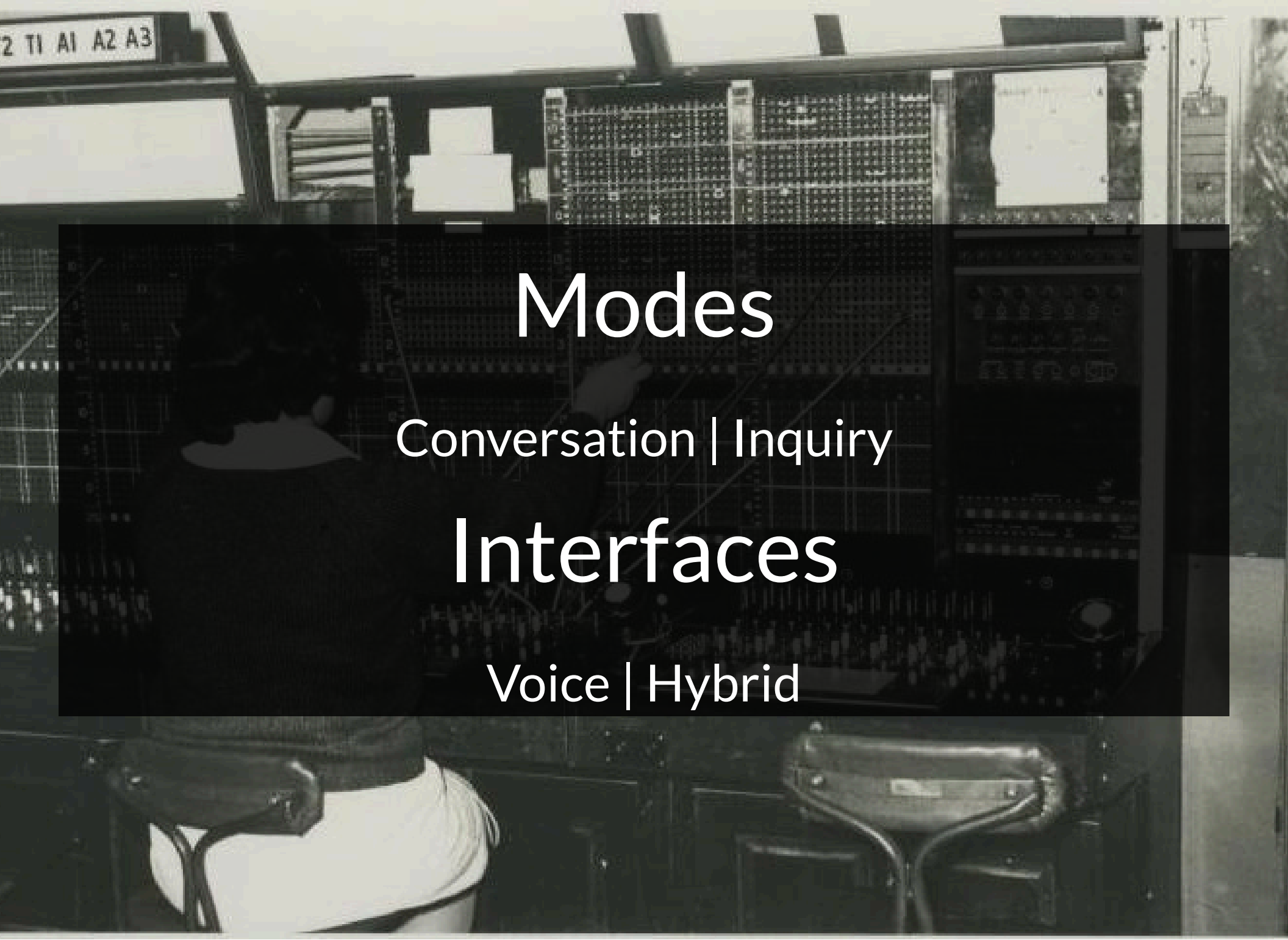


MyStreet

"What's going on around me?"

Chatbots


PRESENT
PAST
FUTURE ?
IVR



Modes

Conversation | Inquiry

Interfaces

Voice | Hybrid

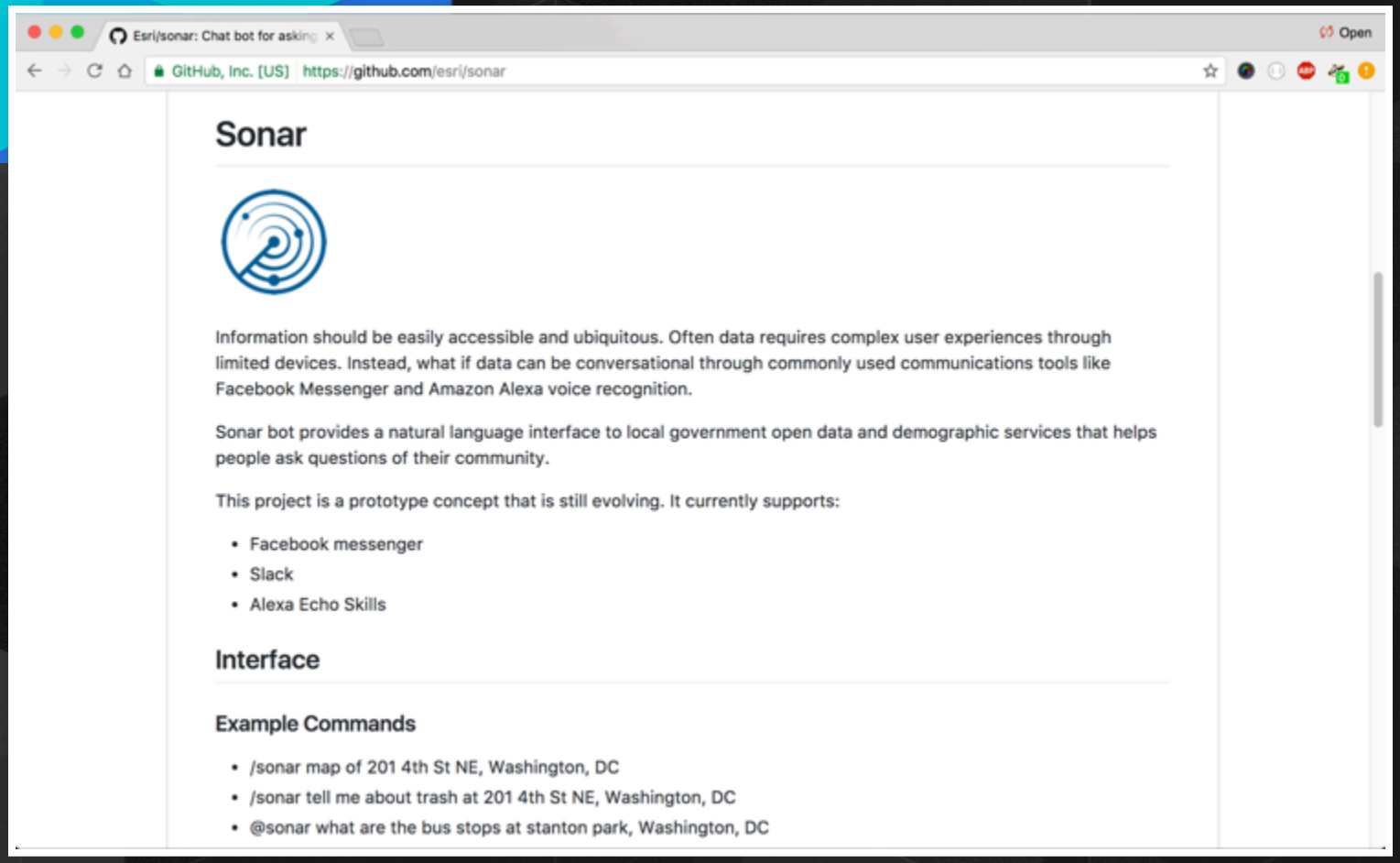
Considerations

- Discoverability vs. Predictability
- Privacy
- Accessibility

'bot'

- Scripted
- AI (aritificial intelligence)

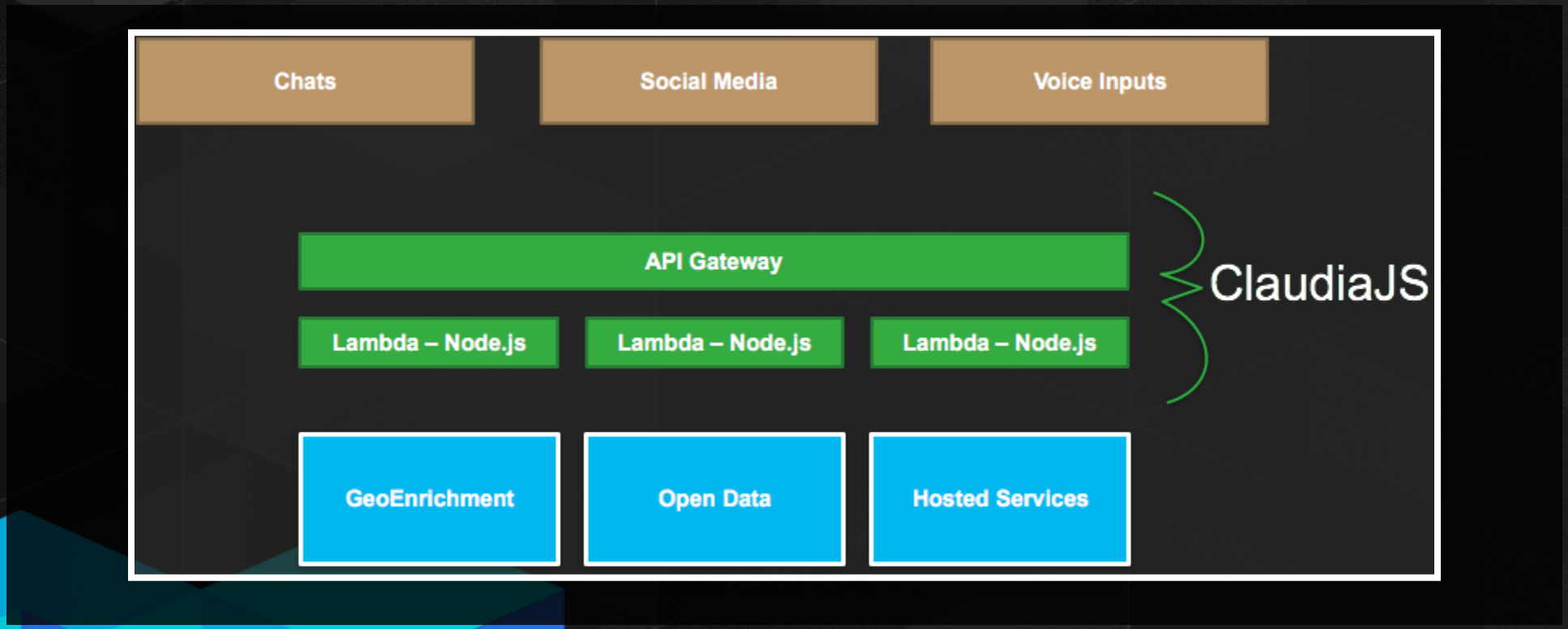
Sonar



Talk to your infrastructure

- Slack
- Facebook fb://sonarhere
- Alexa

Sonar: decomposed


Chats
Social Media
Voice Inputs
API Gateway
ClaudiaJS
Lambda – Node.js
Lambda – Node.js
Lambda – Node.js
GeoEnrichment
Open Data
Hosted Services

Architecture

- Sonar.js
- Claudia.js
- Amazon Web Services
  - Lambda functions
  - API Gateway
  - Alexa skills

ArcGIS

- Geoenrichment
- Geocoding
- Routing

developers.arcgis.com

WebMaps as VUI

Trash

```
Trash pickup is {FREQ_REFUSE} and Recycling is {FREQ_RECYCLING
```

Crime

```
The most recent crime was {OFFENSE}
```



Design for Humans...

- Mon -> Monday
- Tue -> Tuesday
- Thu -> Thursday
- Thurs -> Thursday



experiment

gh://esri/sonar

Andrew Turner

andrew@esri.com | @ajturner

